Unit wise -DBMS Question Bank

Unit I

1. Compare DBMS and File processing systems in terms of data isolation, data redundancy, data inconsistency, data integrity
2. What is DBMS? Explain different applications of DBMS.
3. What is Data abstraction? Explain with suitable diagram
4. Define Key. Explain any two keys
5. Explain DBMS System structure in details
6. Write Short Note On
   i) Database System vs File System
   ii) Three Schema Architecture
   iii) Data Independence
   iv) Types of Data Models
7. A company maintains employee data across multiple departments with shared projects.

i) Identify entities, relationships, and constraints
ii) Convert the ER model into relational schema
iii) Identify weak entities and participation constraints
iv) Discuss design issues and redundancy

8. Explain **Extended Ern Features with suitable diagram in detail.**
9. A reputed general hospital has decided to computerize their operations. In the hospital many doctors are working. Personal information of doctors is maintained to get them fixed salary per month. The patients are admitted to the hospital into the room. They are treated by various doctors. Sometimes patients perform certain pathological tests which are carried out into the labs. Identify the relationship among the entities along with the mapping cardinalities, keys in the E.R. diagram. Construct appropriate tables for E-R diagram designed with above requirements.
10. Draw architecture of DBMS system and explain function of following components: i) Storage manager ii) Query Processor

Unit II

1. Consider following schema Student_fee_details (rollno, name, fee_deposited, date). Write a trigger to preserve old values of student fee details before updating in the table
2. Differentiate between Cursor and Trigger with examples
3. What is the importance of creating constraints on the table? Explain with example any 4 constraints that can be specified when a database table is created.
4. What is view and how to create it? Can you update view? If yes, how? If not, why not?
5. Consider following schema Hotels(hotel_no,hotel_name.city) Rooms(Room_no,hotel_no,price,type). Write a PL/SQL procedure to list the price & type of all rooms at the hotel 'TAJ'
6. Consider the SQL query:

```
SELECT Dept, COUNT(*)
FROM Employee
WHERE Salary > 50000
GROUP BY Dept
HAVING COUNT(*) > 5;
```

1. i) Explain execution steps
   ii) Optimize the query using indexing strategies
   iii) Suggest alternative query formulation
7. Consider the following relation schema MOVIES(Mov_Id, Mov_Title,Mov_Year, Dir_Id) DIRECTOR(Dir_Id, Dir_Name) RATING(MOV_Id, Rev_Stars) Write the SQL queries for the following. i) List the title of all the movies directed by 'RAJ KAPOOR' ii) Find the name of movies and number of stars for each movie. Sort the results on movies title and from higher stars to least stars. iii) Assign the rating of all movies directed by 'Steven Spielberg' to 9.
8. Write SQL queries for the following:

Assume tables:
Student(SID, Name, Dept, Marks)
Course(CID, CName, Credits)
Enroll(SID, CID)

   i) Find names of students scoring more than 75
   ii) Display department-wise average marks
   iii) List students enrolled in more than one course
   iv) Find students not enrolled in any course
9. What is synonym? How to create and use synonym in SQL?
10. Explain: Aggregate Functions ii) Group By and Having Clause
    iii) Nested Queries with example
11. Compare DCL and TCL commands with suitable examples.

Unit III

1. What is Functional dependency. Explain lossless dependency with example
2. Explain normalization with suitable examples: 1NF, 2NF, 3NF and BCNF
3. Given relation:
   R(A, B, C, D, E) FDs: A → B, B → C, A → D, D → E i) Find candidate keys
   ii) Normalize into 3NF
4. What are **anomalies in database design**? Explain insertion, deletion, and update anomalies with example.
5. What is the impact of insert, update and delete anomaly on overall design of database? How normalization is used to remove these anomalies.
6. Elaborate the significance of CODD's rule. Explain 12 rules proposed by CODD's.
7. What is decomposition? Explain the desirable properties of decomposition? Consider the relation F (FN, PN, C, D) with the following Functional Dependencies : FD1: FN, PN ->C FD2: C ->D FD3: D -> F If F is decomposed into Fl(FN, PN ,C) and F2(C, D). Check decomposition is lossless or lossy?
8. Explain Referential Integrity

Unit IV

1. What is Transaction ? Explain with its states
2. What is Schedule and types of schedules?
3. What is conflict serializability? Check following schedule is conflict serializable or not? Also, explain the concept of conflict equivalent schedule.
4.

T1	T2	T3	T4
R(X)			
R(Y)			

	W(X)		
		R(Y)	
		W(Y)	
			W(X)
			W(Y)
			W(Z)

R(X) denotes read operation on data item X by transaction Ti . W(X) denotes write operation on data item X by transaction Ti .

5. What is recoverable schedule? Why is recoverability of schedule desirable? Are there any circumstance under Which it could be desirable to allow non recoverable schedular? Explain your answer.
6. What are ACID Properties? Explain
7. How to ensure atomicity using Recovery Methods? Explain the log based recovery method in detail.
8. Check whether the given schedule is conflict-serializable:
   T1: R(A), W(A)
   T2: R(A), W(A)
   Construct precedence graph.
9. Differentiate between View and Conflict Serializable
10. Explain serializability with examples.
11. Explain concurrency control techniques: i) Lock-based protocols ii) Two Phase Locking (2PL) iii) Deadlock handling
12. Explain **View Serializability vs Conflict Serializability** with examples where conflict serializability fails but view serializability holds.
13. State and explain the ACID properties. During its execution a transaction passes through several states, until it finally commits or aborts. List all possible sequences of states through which a transaction may pass. Explain the situations when each state transition occurs
14. Explain Cascade less Rollback with Dirty Read

Unit V

1. What is NoSQL. What are applications?
2. Differentiate between SQL and NoSQL
3. Explain the NOSQL database types with examples and write down the real time applications
4. List the different NOSQL data models. Explain following NOSQL database types with examples. i) Column-oriented ii) Document-oriented
5. Explain the CAP theorem referred during the development of any distributed application
6. Describe the following operations with MongoDB syntax : i) Map Reduce ii)Aggregation pipeline
7. Explain MongoDB features and operations with examples: i) CRUD ii) Indexing iii) Aggregation
8. Explain NoSQL databases with advantages and limitations.
9. Solve the following counting problems:
   Explain NoSQL databases:
   i) Key-Value Model ii) Document Model iii) Column Family iv) Graph Database